

Final Report: Modeling Kinetics of Bacteriophage in Phage Therapy

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Introduction

Phage Therapy is the use of viruses (known as bacteriophages or phages) that specifically target certain bacteria and can then be used to treat bacterial infections. The clinical application of phage therapy is very promising, but the outcome is heavily dependent on various conditions throughout the treatment. You can model the relationship between bacteria and phages in many ways using models such as DDE (delay differential equations) or ODE (ordinary differential equations). I will focus on an ODE representation of this relationship throughout the project and report.

Methods

I decided to use the following equations obtained from [this study](#) to model this relationship in C++ using the Runge-Kutta ODE approximation:

$$\frac{dx}{dt} = ax - bvx \quad X(t) \text{ represents the number of uninfected bacteria,}$$

$$\frac{dy}{dt} = ay + bvx - ky \quad Y(t) \text{ represents the number of infected bacteria,}$$

$$\frac{dv}{dt} = kLy - bvx - mv \quad V(t) \text{ represents the free phage.}$$

Where a is the replication coefficient of bacteria, b is the transmission coefficient, k is the lysis rate, L is the burst size, and m is the decay rate of the free phages.

Tests/Verification/Validation

I tested my program by comparing it to a chart that was mentioned in the study, which was generated using the same ODEs that I have implemented. The given chart can be found in Figure 1:

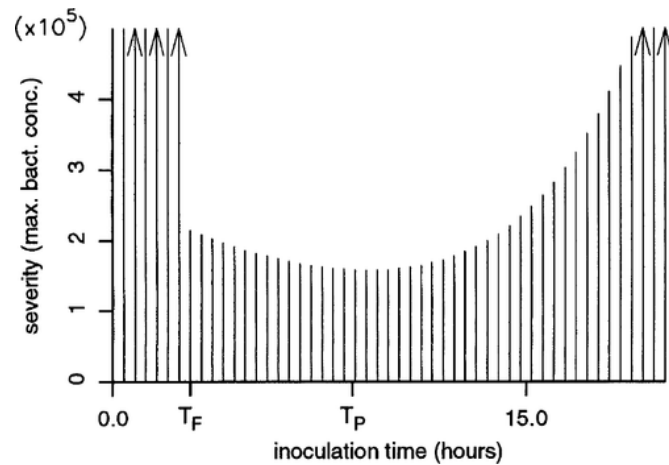


Figure 1 (Graph obtained from study 1.)

This chart is generated by using the following parameters (as mentioned in the study): $a = 0.3$, $b = 10^{-6}$, $k = 1.2$, $L = 100$, and $m = 1.8$. The issue I came across with using these values, is that the study did not provide me with sufficient information on what the initial values of x and y (the uninfected and infected bacteria) were for this study (initial phage bacteria is mentioned and is 100). I then searched online for other studies on Phage Therapy and bacteriophages in general to get a rough estimate of 3×10^4 uninfected bacteria and 1×10^4 infected bacteria. I then used these values to generate the following graph:

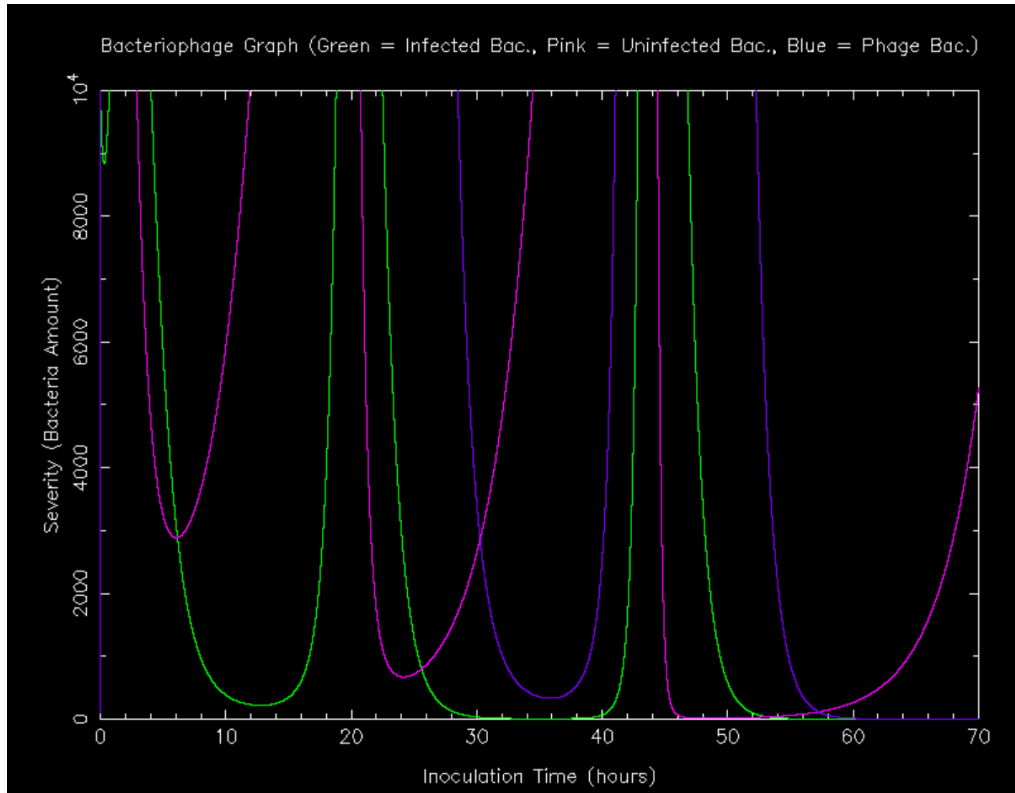


Figure 2

Graph generated using `phageTherapy.cpp` code with parameters

$$a = 0.3, b = 10^{-6}, k = 1.2, L = 100, m = 1.8 \text{ with initial values } x = 3 \times 10^4, y = 1 \times 10^4, v = 100$$

Comparing this graph and Figure 1, we can observe that they are different graphs. This occurs since the graph in the study is not a typical line graph and the initial values are unknown. Even with these discrepancies, we can still pull the general pattern from both graphs that only in certain inoculation time intervals will the phage bacteria reduce the infected bacteria significantly. In Figure 1 we can see that the optimal inoculation time is around 9 hours, while in Figure 2 the optimal inoculation time (for the highest number of uninfected bacteria) would be around 11 hours. Given that we do not know the initial values for Figure 1, the two-hour difference between these graphs makes sense as the Figure 2 bacteria counts are only a rough estimate of what Figure 1 uses. We can verify that these results make sense by

using another set of initial bacteria values with the same initial phage bacteria values. Using the initial values of 1×10^3 and 3×10^3 for infected and uninfected bacteria, we get the following graph:

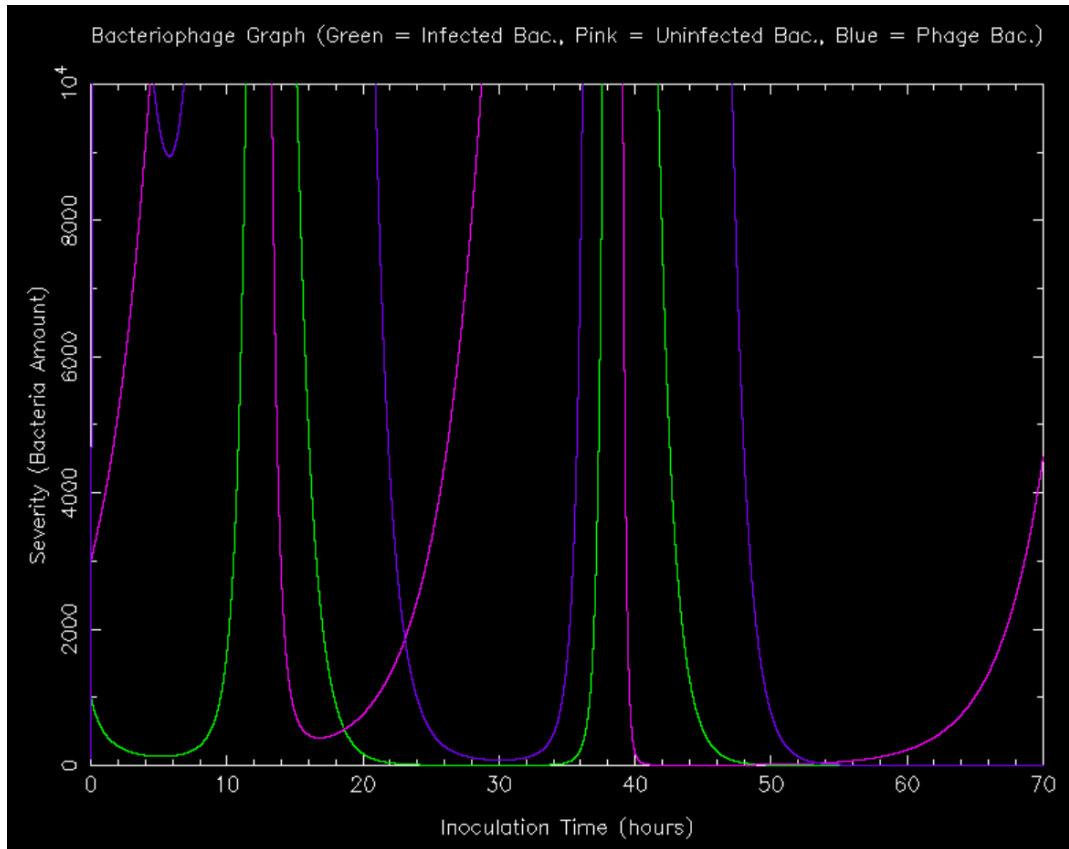


Figure 3

Graph generated using `phageTherapy.cpp` code with parameters

$$a = 0.3, b = 10^{-6}, k = 1.2, L = 100, m = 1.8 \text{ with initial values } x = 3 \times 10^3, y = 1 \times 10^3, v = 100$$

Figure 3 shares the same patterns as Figure 1/2, where only certain intervals have the phages reduce the infected bacteria. Although the intervals are rather different than both Figures 1 and 2, they still represent the same pattern.

Results

Once I verified that my graph was correctly implementing the Runge-Kutta method with the 3 ODEs from the study, I then decided to explore how the relationship between infections and inoculation time changed when modifying/adjusting the parameters.

When modifying the parameters used in Figure 2 from $a = 0.3$ to $a = 0.8$, we get the following graph:

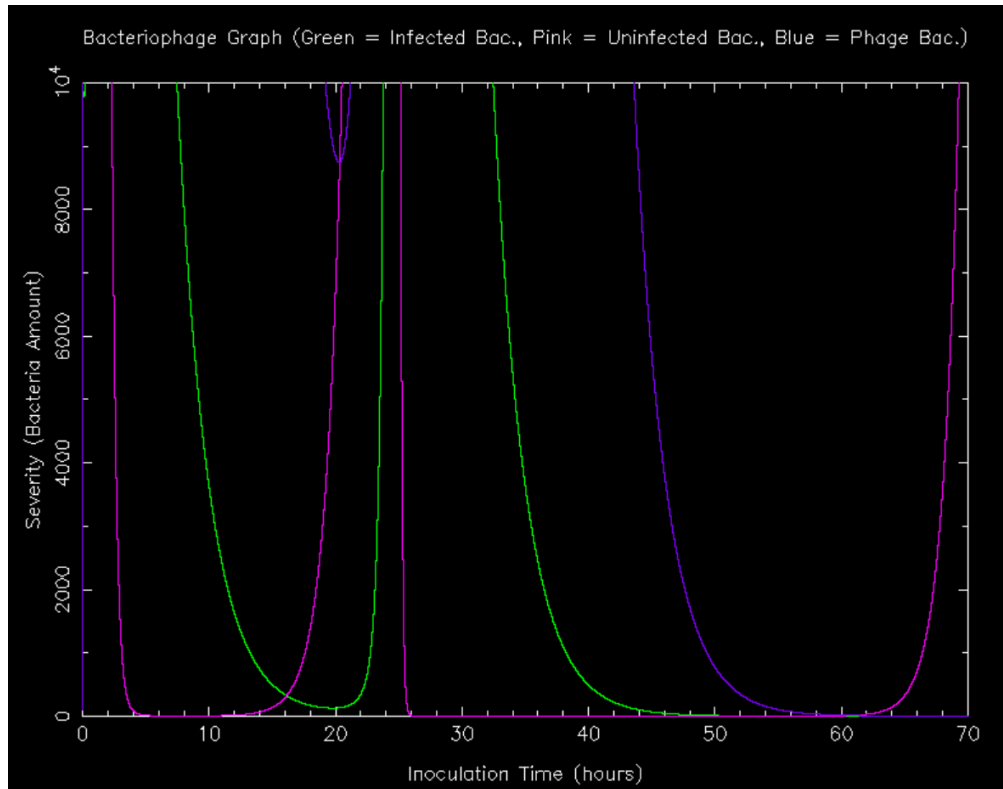


Figure 4

Graph generated using phageTherapy.cpp code with parameters

$a = 0.8$, $b = 10^{-6}$, $k = 1.2$, $L = 100$, $m = 1.8$ with initial values $x = 3 \times 10^4$, $y = 1 \times 10^4$, $v = 100$

Comparing Figure 4 and Figure 2, we can observe that increasing the a parameter value resulted in higher spikes of bacteria numbers for all infected/uninfected/phage bacteria. This observation makes sense since a is the replication coefficient for both the infected and uninfected bacteria in the system, therefore increasing the coefficient should result in higher bacteria concentrations/spikes for

infected/uninfected bacteria. Since the phage bacteria values depend on the infected/uninfected bacteria, the phage concentrations will also be higher as a result.

When changing the transmission coefficient b in Figure 2 from 10^{-6} to 10^{-3} , we will get the following graph:

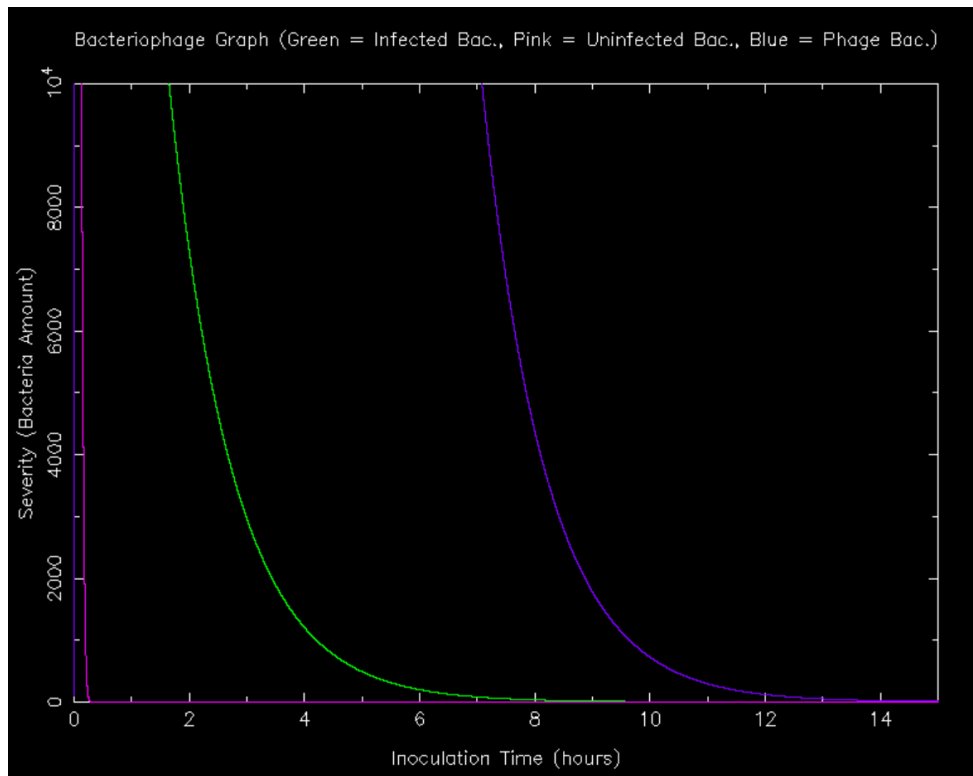


Figure 5

Graph generated using `phageTherapy.cpp` code with parameters

$$a = 0.3, b = 10^{-3}, k = 1.2, L = 100, m = 1.8 \text{ with initial values } x = 3 \times 10^4, y = 1 \times 10^4, v = 100$$

This graph makes sense as a higher transmission coefficient means that more uninfected bacteria will become infected, and therefore their bacteria population would drop very quickly as can be seen from

the pink line in Figure 5. Since there are no more uninfected bacteria, both the infected and phage bacteria will slowly die out.

When changing the lysis rate coefficient k from 1.2 to 1.8, we get the following graph:

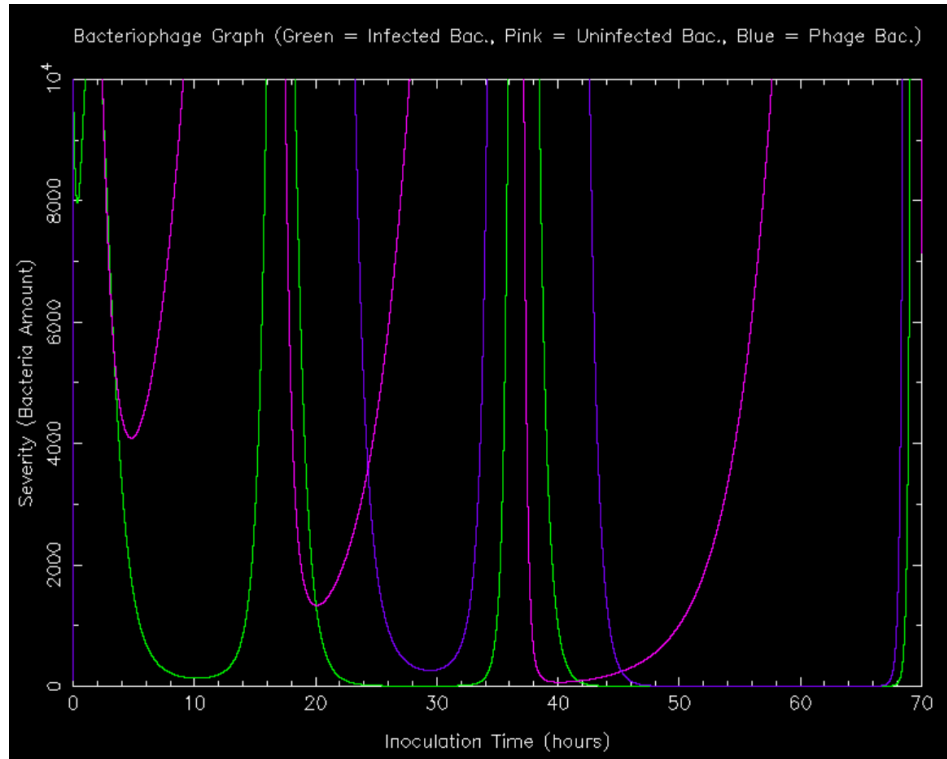


Figure 6 (Graph generated using phageTherapy.cpp code with parameters

$a = 0.3$, $b = 10^{-6}$, $k = 1.8$, $L = 100$, $m = 1.8$ with initial values $x = 3 \times 10^4$, $y = 1 \times 10^4$, $v = 100$)

Comparing Figure 2 and Figure 6, we can see the relationship is that when the lysis rate is increased, the overall inoculation time for each bacterium is speed-up/shortened. All the lines on the graph have the same curve/pattern but they happened at earlier hours in Figure 6 than in Figure 2. This makes sense because if the lysis rate is larger that means the bacteria in the system will break down quicker and therefore will have a shorter lifespan.

When changing the burst size L from 100 to 200, we get the following graph:

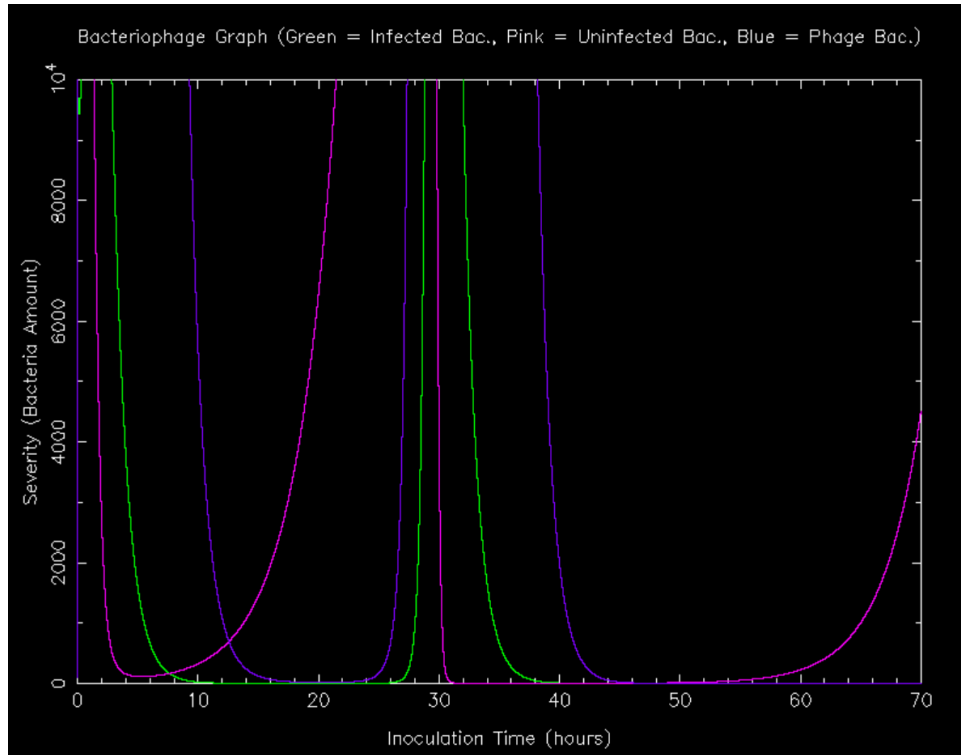


Figure 7 (Graph generated using *phageTherapy.cpp* code with parameters

$$a = 0.3, b = 10^{-6}, k = 1.2, L = 200, m = 1.8 \text{ with initial values } x = 3 \times 10^4, y = 1 \times 10^4, v = 100)$$

By comparing Figure 2 and Figure 7, we can observe the relationship that the larger the burst size is, the lower (and for longer) the initial drop for the infected, uninfected, and phage bacteria is. This makes sense as a larger burst means more phage bacteria, and therefore quicker reduction in infected bacteria.

When changing the decay rate of the free phage m from 1.8 to 2.5, we get the following graph:

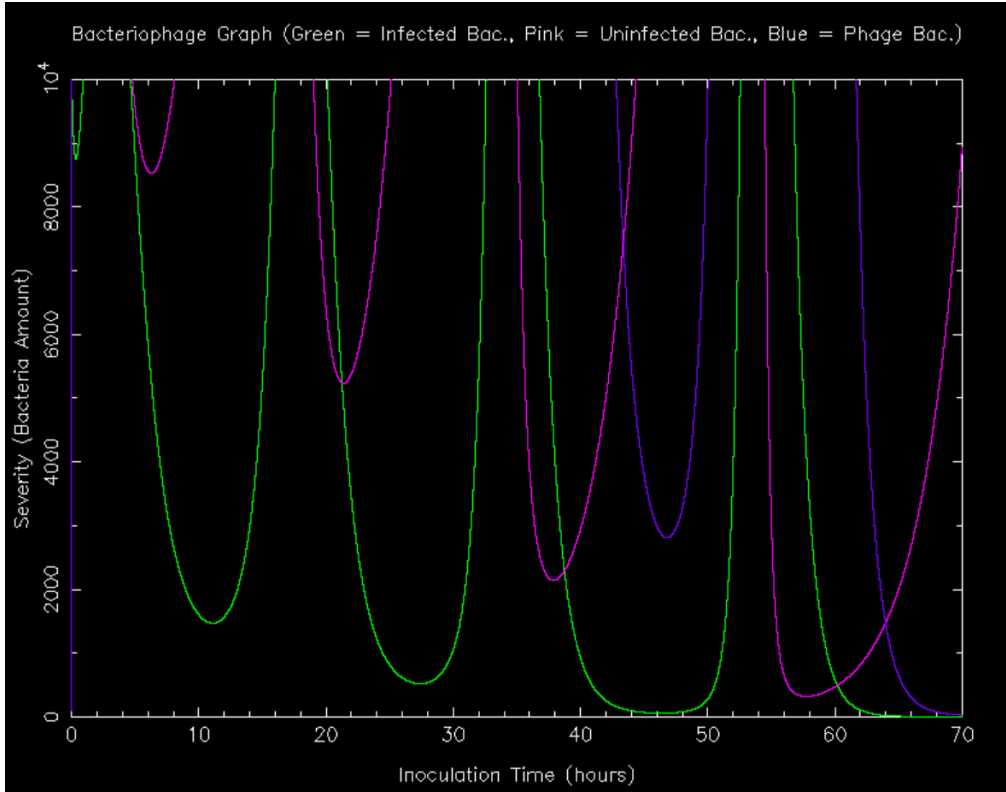


Figure 8 (Graph generated using *phageTherapy.cpp* code with parameters

$$a = 0.3, b = 10^{-6}, k = 1.2, L = 100, m = 2.5 \text{ with initial values } x = 3 \times 10^4, y = 1 \times 10^4, v = 100)$$

By comparing Figure 2 and Figure 8, we can see that if we increase the decay of the free phage, the bacteria counts do not hit as low values as before. This makes sense because having a higher decay rate means there will be smaller amounts of phage bacteria in the system and since the uninfected bacteria ODE is $ax - bvx$, there will be a larger number of uninfected bacteria. Since there is a larger number of uninfected bacteria, there will be a larger number of infected bacteria as they grow in correspondence to the uninfected bacteria quantity. The phage bacteria don't drop as much because they rely on the infected/uninfected bacteria values too.

Conclusions

Using the ODEs from [this study](#), I was able to model the kinetics of bacteriophages with infected and uninfected bacteria by implementing the Runge-Kutta ODE approximation method in C++ and graphing the results using PGPLOT. Using these results, I was able to establish that the inoculation time is a very important aspect in the effectiveness of bacteriophages throughout its real-world applications such as Phage Therapy, and that there are many other factors that are important like the replication coefficient, lysis rate, or transmission coefficient.

References

1. <https://ascpt.onlinelibrary.wiley.com/doi/full/10.1067/mcp.2000.109520>
2. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8490754/> (used to understand what initial values should be)

Code

phageTherapy.cpp

```
#include <iostream>
#include <cmath>
#include "cpgplot.h"

double dxdt(double x, double v, double a, double b) {
    return (a * x) - (b * v * x);
}

double dydt(double y, double v, double x, double a, double b, double k) {
    return (a * y) + (b * v * x) - (k * y);
}

double dvdt(double v, double y, double x, double k, double L, double m, double b)
{
    return (k * L * y) - (b * v * x) - (m * v);
}

// rk4(a, b, k, L, m, initial x value, initial y value, initial v value, final
inoculation time)
```

```

int rk4(double a, double b, double k, double L, double m, double x, double y,
double v, int tend) {
    //initialize variables
    int n;
    double tstart;
    double t;
    double kx1 = 0, kx2 = 0, kx3 = 0, kx4 = 0;
    double ky1 = 0, ky2 = 0, ky3 = 0, ky4 = 0;
    double kv1 = 0, kv2 = 0, kv3 = 0, kv4 = 0;
    tstart = 0; //start at 0
    n = 1000000; //set n to 1000000
    double deltaT = (tend - tstart)/n;
    t = tstart;
    int i = 0;
    float tp[n+1];
    float yp[n+1];
    float xp[n+1];
    float vp[n+1];
    tp[0] = t;
    yp[0] = y;
    xp[0] = x;
    vp[0] = v;
    //rk4 estimation Loop
    while (i < n) {
        //k1
        kx1 = deltaT * (dxdt(x, v, a, b));
        ky1 = deltaT * (dydt(y, v, x, a, b, k));
        kv1 = deltaT * (dvdt(v, y, x, k, L, m, b));
        //k2
        kx2 = deltaT * (dxdt(x + 0.5 * kx1, v + 0.5 * kv1, a, b));
        ky2 = deltaT * (dydt(y + 0.5 * ky1, v + 0.5 * kv1, x + 0.5 * kx1, a, b,
k));
        kv2 = deltaT * (dvdt(v + 0.5 * kv1, y + 0.5 * ky1, x + 0.5 * kx1, k, L,
m, b));
        //k3
        kx3 = deltaT * (dxdt(x + 0.5 * kx2, v + 0.5 * kv2, a, b));
        ky3 = deltaT * (dydt(y + 0.5 * ky2, v + 0.5 * kv2, x + 0.5 * kx2, a, b,
k));
        kv3 = deltaT * (dvdt(v + 0.5 * kv2, y + 0.5 * ky2, x + 0.5 * kx2, k, L,
m, b));
        //k4
        kx4 = deltaT * (dxdt(x + kx3, v + kv3, a, b));
        ky4 = deltaT * (dydt(y + ky3, v + kv3, x + kx3, a, b, k));
        kv4 = deltaT * (dvdt(v + kv3, y + ky3, x + kx3, k, L, m, b));
        //update values

```

```

        t = t + deltaT;
        x = x + (1.0/6.0) * (kx1 + 2*kx2 + 2*kx3 + kx4);
        y = y + (1.0/6.0) * (ky1 + 2*ky2 + 2*ky3 + ky4);
        v = v + (1.0/6.0) * (kv1 + 2*kv2 + 2*kv3 + kv4);
        i++;
        tp[i] = t;
        yp[i] = y;
        xp[i] = x;
        vp[i] = v;
    }
    // Graph Results
    if (!cpgopen("/XWINDOW")) return 1;
    cpgenv(tstart, tend, 0, 10000, 0, 0);
    cpglab("Inoculation Time (hours)", "Severity (Bacteria Amount)",
"Bacteriophage Graph (Green = Infected Bac., Pink = Uninfected Bac., Blue = Phage
Bac.)");
    cpgsci(3); //green line = y = infected bac.
    cpgline(n + 1, tp, yp);
    cpgsci(6); //pink line = x = uninfected bac.
    cpgline(n + 1, tp, xp);
    cpgsci(12); //purple line = v = phage bac.
    cpgline(n + 1, tp, vp);
    cpgclos();
    return 0;
}

int main() {
    // default parameters given from study
    double a = 0.3;
    double b = pow(10, (-6));
    double k = 1.2;
    double L = 100;
    double m = 1.8;

    // Initial Values For X, Y, V
    double x = 3 * pow(10, 4); // uninfected bacteria number
    double y = 1 * pow(10, 4); // infected bacteria number
    double v = 100; //phage bacteria number

    // Tests

    // Demo Test
    rk4(a, b, k, L, m, x, y, v, 70);

```

```

    // Other Results Tests
    //rk4(0.8, b, k, L, m, x, y, v, 70);
    //rk4(a, pow(10, (-4)), k, L, m, x, y, v, 15);
    //rk4(a, b, 1.8, L, m, x, y, v, 70);
    //rk4(a, b, k, 300, m, x, y, v, 70);
    //rk4(a, b, k, L, 2.5, x, y, v, 70);
    return 0;
}

```

Makefile

```

phageTherapy: phageTherapy.o
    c++ -o phageTherapy phageTherapy.o -ltrapfpe -lpgplot -lcpgplot -lX11 -lm

phageTherapy.o: phageTherapy.cpp
    c++ -c phageTherapy.cpp

```